

NAKISKA RIDGETOP, AB, Canada

WMO: 712450

Lat: 50.9437N

Lon: 115.1896W

Elev: 8343

StdP: 10.77

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.2	-11.3	-26.4	1.7	-9.0	-20.0	2.5	-1.5	73.1	22.8	61.9	21.5	8.1	50	1.058

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	65.1	48.1	62.2	47.0	59.5	45.8	51.2	59.6	49.2	58.4	47.6	57.0	12.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
47.9	67.7	53.1	45.3	61.2	51.9	43.0	55.9	50.3	24.1	60.0	22.8	58.4	21.7	57.2	60.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	52.7	43.3	-25.6	70.6	6.7	3.3	-30.4	73.0	-34.3	75.0	-38.0	76.9	-42.9	79.3
			-24.3	54.4	7.3	2.5	-29.5	56.1	-33.7	57.6	-37.8	59.0	-43.1	60.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.3	16.1	14.3	18.4	25.1	34.3	40.7	49.1	48.0	40.2	29.0	20.1	15.2
	DBStd	15.32	10.77	10.64	9.51	8.26	7.98	6.86	7.34	7.33	9.51	9.08	9.92	10.31
	HDD50	7736	1051	999	979	746	490	291	106	127	320	651	898	1078
	HDD65	13027	1516	1419	1444	1196	952	728	493	528	745	1115	1348	1543
	CDD50	184	0	0	0	0	2	13	79	64	25	1	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	16.4	21.6	16.0	19.2	14.6	13.2	12.5	12.9	11.9	14.8	17.7	21.5	20.9
Precipitation	PrecAvg	25.7	2.1	1.4	1.7	1.5	2.4	3.5	2.4	2.3	2.1	2.0	2.3	2.0
	PrecMax	33.7	3.3	2.2	3.0	3.3	4.1	7.9	4.6	4.4	4.1	3.7	5.4	3.1
	PrecMin	20.0	0.6	0.5	0.5	0.7	1.1	2.1	1.0	0.6	0.5	1.1	1.0	1.1
	PrecStd	2.7	0.6	0.5	0.6	0.6	0.7	1.3	0.9	0.9	0.9	0.6	1.0	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.0	37.1	41.5	49.6	57.1	64.0	70.2	68.3	64.7	54.6	43.8	41.7
		MCWB	27.9	27.2	29.3	37.0	42.0	47.4	50.5	48.9	45.8	39.7	32.4	29.2
	2%	DB	35.2	33.6	35.6	44.1	52.9	58.6	66.1	64.1	60.4	48.8	39.4	36.2
		MCWB	25.6	26.2	27.7	34.0	39.9	45.8	49.1	47.8	44.5	36.2	30.9	26.6
	5%	DB	31.7	30.0	32.4	40.2	49.5	54.6	63.0	61.3	57.2	45.1	36.1	31.7
		MCWB	25.1	24.6	27.1	32.0	38.2	43.9	47.8	46.5	43.1	35.1	30.2	25.3
	10%	DB	28.8	27.2	29.6	36.8	46.1	50.9	60.0	58.6	54.0	42.0	32.7	28.4
		MCWB	24.0	23.2	25.4	30.3	36.7	41.6	46.6	45.6	41.8	33.4	27.6	23.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.5	29.8	31.7	37.6	43.9	51.2	54.9	52.8	49.1	41.9	35.0	31.1
		MCDB	38.8	34.4	38.0	47.2	54.5	58.5	64.1	59.9	60.1	51.1	39.6	37.8
	2%	WB	27.9	26.9	29.5	35.2	41.5	47.8	52.3	50.2	46.3	38.5	32.5	28.3
		MCDB	32.9	31.7	34.4	43.0	51.0	56.9	60.6	59.4	57.4	46.5	36.9	34.7
	5%	WB	25.8	25.0	27.7	33.1	39.5	45.2	50.0	48.6	44.5	36.0	29.9	25.9
		MCDB	29.8	29.1	32.4	39.5	48.2	54.1	58.6	57.7	55.5	43.3	34.2	31.5
	10%	WB	23.8	22.7	25.7	30.8	37.5	43.1	48.2	47.0	42.6	34.0	27.2	23.7
		MCDB	27.7	26.7	29.9	36.2	45.6	50.7	57.6	56.4	53.0	41.3	31.1	28.1
Mean Daily Temperature Range	5% DB	MDBR	9.9	10.3	10.0	10.7	10.5	10.7	12.8	12.5	11.0	9.8	8.9	9.3
		MCDBR	10.5	10.6	11.2	13.3	13.1	14.2	15.1	14.7	13.8	12.0	10.2	11.1
		MCWBR	7.2	6.9	8.2	7.7	6.9	7.9	7.5	7.2	6.9	7.3	7.3	8.0
	5% WB	MCDBR	9.8	9.8	11.0	12.5	11.9	13.3	14.9	14.5	13.6	12.4	9.6	10.7
		MCWBR	7.7	7.9	8.7	8.3	7.1	8.0	8.6	8.1	7.4	8.2	7.9	8.5
Clear-Sky Solar Irradiance	taub	0.206	0.217	0.234	0.270	0.280	0.291	0.285	0.293	0.264	0.237	0.214	0.199	
	taud	2.490	2.492	2.486	2.408	2.513	2.519	2.553	2.539	2.612	2.675	2.586	2.478	
	Ebn at Noon	283	306	316	311	310	306	307	299	300	295	279	270	
	Edh at Noon	16	22	27	33	31	31	30	29	24	18	15	14	
All-Sky Solar Radiation	RadAvg	354	656	1029	1443	1676	1770	2000	1660	1167	725	383	280	
	RadStd	23	44	70	99	141	147	147	119	133	71	26	17	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page